

DuHa: a dual-hand action segmentation method for human-robot collaborative assembly

Hao Zheng*, Regina Lee, Yuqian Lu, Xun Xu

Abstract—Human-robot collaboration (HRC) is crucial for enabling mass personalised manufacturing and human-centric manufacturing. The recent advancements in video understanding technology have enabled robots to interpret human actions from videos and discern the appropriate timing and nature of required robot assistance. However, current vision-based HRC systems mainly focus on capturing human overall actions, neglecting the detailed actions of each hand. This restricts robots from providing finer attentive assistance, such as delivering the required tools to a specific hand. Furthermore, these systems predominantly utilise classification-based action recognition techniques rather than reasoning-involved action segmentation techniques, thereby constraining their practical application. To address above limitations, this paper proposes a vision-based dual-hand action segmentation method for human-robot collaborative assembly (HRCA) scenarios. Initially, graph neural networks are used to capture hand-object interactions for both hands across frames, then these features are integrated with scene features, and finally, temporal convolutional networks are applied to capture temporal dynamics of the integrated features over frames. To better serve the application needs of HRCA, we have refined the action segmentation to atomic action level according to the Human-Robot Shared Assembly Taxonomy, facilitating comprehensive collaboration. More details can be found at <https://github.com/LISMS-AKL-NZ/DuHa>

I. INTRODUCTION

Human-robot collaboration (HRC) is integral to the paradigm of mass personalised manufacturing, synergistically combining human cognitive advantages with robotic efficiency [1], [2]. This collaboration can reduce human physical and cognitive load through timely robotic assistance, which aligns with the principles of human-centric manufacturing [3], [4]. Vision-based HRC systems, compared to other approaches, are more intuitive, as they enable robots to offer contextualised assistance based on visual observations alone, without the need for supplementary sensors.

Recent progress in video understanding has markedly advanced vision-based HRC systems, shifting their focus from static image analysis [5], [6] to the dynamic interpretation of human actions [7], [8]. Real-time and cumulative human action understanding allows robots to discern human intentions and task progression, so as to optimise task allocation [9], [10] and deliver personalised assistance [11]. However, existing vision-based HRC systems typically provide only a broad

overview of human actions, lacking the granularity needed for understanding individual hand actions. This limitation becomes particularly evident in product assembly scenarios, where two-hand collaborative and parallel tasks are common, thus hindering robots' capacity for nuanced task allocation and finer attentive assistance planning. Another technical limitation of existing vision-based HRC systems is that they predominantly employ action recognition techniques, which predict an action label for a given video clip [12], rather than action segmentation techniques, which segment all action instances in a long video [13]. In HRC contexts, the primary objective of action understanding should be accurate identification and tracking of human actions. Therefore, action segmentation, which refines frame-wise action predictions by considering the context of preceding and subsequent actions, better fulfils the HRC requirements. Driven by these insights, we propose a dual-hand action segmentation method (DuHa) for human-robot collaborative assembly (HRCA) scenarios.

Existing action segmentation methods focus on extracting spatiotemporal features of each frame and modelling the temporal dynamics of these features across frames [14], [15], [16], [17]. These spatiotemporal features capture the motion information in a scene, which is essential for general action understanding. However, for assembly action understanding, we believe that the interaction state changes between two hands and other objects are crucial. Therefore, in our proposed method, we initially utilise graph neural networks (GNN) to capture the hand-object interactions for both hands across frames. Subsequently, the hand-object interaction feature of each hand is integrated with the general spatiotemporal scene feature at each frame. Finally, we leverage temporal convolutional networks (TCN) to capture the temporal dynamics of the integrated features over frames. This approach not only augments feature representation for assembly scenarios but also enables synchronously parsing dual-hand actions using a single model.

Assembly action understanding in industrial contexts is notably challenging, primarily because of 1) the vast array of assembly actions, 2) the prevalence of actions with subtle motions, 3) the numerous analogous actions, and 4) the common issue of occlusion in videos. Existing vision-based HRCA systems [18], [19] have been validated primarily in simplistic scenarios, characterised by a limited range of actions and low complexity in action sequences. In this research, we evaluated DuHa on a more challenging dataset, HA-ViD (a human assembly video dataset) [20], where experimental results demonstrate the superiority of DuHa over the existing algorithms.

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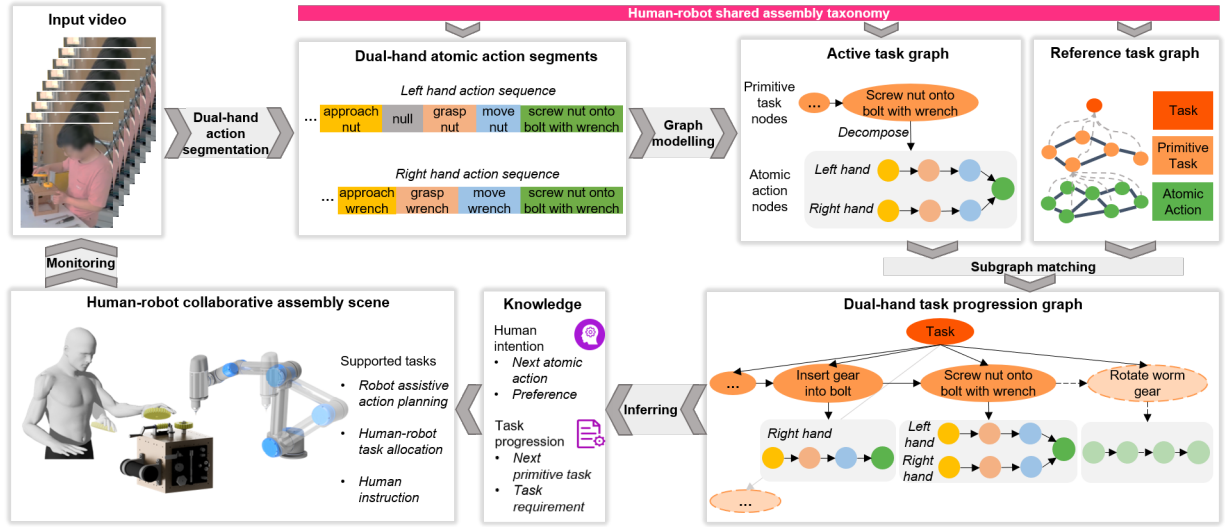


Fig. 1. Dual-hand action segmentation-based human-robot collaborative assembly framework.

According to Human-Robot Shared Assembly Taxonomy (HR-SAT) [21], an assembly task can be decomposed into a series of primitive tasks, and further into a series of atomic actions. HRCA may take place at primitive task level, such as primitive task allocation, or at atomic action level, such as robots delivering tools to humans. We conducted action segmentation at atomic action level for bottom-up task understanding, thereby enabling robot assistance planning at both levels. HR-SAT provides a standardized, objective method for describing and decomposing assembly tasks, adaptable to various scenarios. The dual-hand action annotations in HA-ViD adhere to HR-SAT, enabling DuHa, trained on HA-ViD, to gain transferability.

This paper is organised as follows. Section 2 provides a brief review of research on human-robot collaborative assembly and vision-based action understanding. Section 3 presents a HRCA framework and outlines the method for the proposed DuHa. Section 4 introduces the experimental results of our method. Lastly, Section 5 concludes this paper.

II. RELATED WORK

A. Human-robot collaborative assembly

HRCA enhances efficiency by harnessing the synergistic collaboration between humans and robots [22]. The organizational psychology suggests that the shared understanding of the task and each individual's role is key to effective collaboration [23]. In HRCA, the prerequisite technical challenge lies in enabling robots to understand the shared task, environment and human intentions. The successful integration of computer vision technology in industry underscores its significance in perception [24], [25], [26] and establishes it as an important method in assembly environment perception for HRCA. Recent progress in video understanding technology has propelled HRCA towards human action understanding, as the actions are key indicators of human intentions and task progression. To enhance the reliability of action understanding, various studies have explored the integration

of additional sensors [27], [18]. Nevertheless, extra sensors also impose practical constraints, underscoring a trade-off between complexity and utility.

B. Vision-based action understanding

Recent years have witnessed the flourish of vision-based action understanding techniques, especially in action recognition, which has made impressive progress under various models [12]. However, the current research trend is shifting from merely classifying a single action for a short video clip to segmenting a series of actions in a longer video sequence [28]. We believe that action segmentation is better aligned with the objectives of action understanding in HRCA. Current action segmentation approaches commonly employ temporal networks on the pre-extracted spatiotemporal scene features to capture the features' dependencies across the entire video for accurate frame-wise action classification [14], [15], [16], [17], [29]. However, these motion features offer only a general description on the scene motion and lack the ability to represent the hand-object interactions, which are essential for dual-hand action understanding in assembly contexts. Therefore, assembly action segmentation should consider integrating human-object interaction features, where Graph Neural Networks (GNN) have shown promising performance in learning these features [30].

III. METHOD

A. Conceptual framework of a dual-hand action segmentation based HRCA

The main contribution of this paper is the development of a dual-hand action segmentation method for HRCA scenarios. Before detailing this method, we propose a framework to illustrate its integration into a HRCA system (see Fig. 1). Within it, dual-hand actions are initially segmented from the input video, and subsequently modelled as an active task graph (ATG) depicting historical action types, durations and sequences. Following the task decomposition protocol

of HR-SAT [21], we conduct action segmentation at atomic action level. With HR-SAT, ATGs are generated for both atomic actions and primitive tasks. The ATG is then aligned with the reference task graph (a predefined primitive task precedence graph) to track the overall task progression. Finally, we can infer subsequent tasks and human actions, thereby enabling robots to offer comprehensive assistance, including robot assistive action planning, human-robot task allocation, and human assembly instruction.

B. Daul-hand action segmentation

We propose DuHa for dual-hand action segmentation, as shown in Fig. 2. In DuHa, we utilise a scene feature extractor to derive frame-wise overall motion features, denoted as

$$X_{sce} = \{x_{sce,1}, x_{sce,2}, \dots, x_{sce,T}\}$$

where T is the total number of frames. Parallely, the video is represented as a hand-centric graph

$$g = \{G_1, G_2, \dots, G_T\}$$

using the detected object bounding boxes. $G_t = (V_t, E_t)$ represents the frame at time t . The hand-centric graph includes nodes for left hand $v_{lh,t}$, right hand $v_{rh,t}$, and objects $V_{obj,t}$, with hand-object edges

$$E_t = \{e_{hand-obj,t} \mid e_{hand-obj,t} \text{ connects } v_{hand,t} \text{ and } v_{obj,t} \text{ for } hand \in \{lh, rh\} \text{ and each object } obj\}.$$

By exclusively incorporating hand-object edges, the hand-centric graph more effectively and directly models the relationships between both hands and objects. Node features are represented as

$$x_{i,t} = [\text{loc}_{i,t}, \text{emb}_{i,t}]$$

where $i \in V_t$, $\text{loc}_{i,t}$ and $\text{emb}_{i,t}$ are the location and class embedding for node i at frame t , respectively.

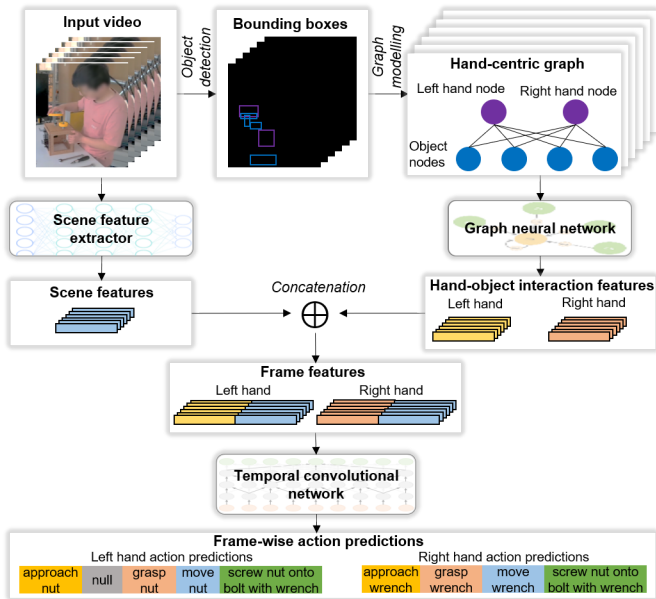


Fig. 2. Overview of DuHa network.

a) *Missing node compensation*: Occlusion, a prevalent problem in videos, manifests as missing nodes in our graph representation. To mitigate this, we design a missing node compensation mechanism. Missing nodes are characterised by their temporary disappearance and subsequent reappearance in a video. For a missing node $v_{i,t}$ at frame t , we estimate its location by interpolating between the last visible location of the node at frame t_{last} , and its next visible location at frame t_{next} . The interpolated location for the missing node at frame t is calculated as follows:

$$\text{loc}_{i,t} = \text{Interpolate}(\text{loc}_{i,t_{last}}, \text{loc}_{i,t_{next}}, t, t_{last}, t_{next}) \quad (1)$$

Then the compensated missing node feature $x_{i,t}$ is obtained by integrating the interpolated location $\text{loc}_{i,t}$ and class embedding $\text{emb}_{i,t}$.

b) *Spatial propagation*: In this phase, our objective is to learn features that capture hand-object interactions per frame while integrating the frame's overall scene feature.

First, the Graph Attention Network (GAT) [31] is used to update the features for each hand and object in each frame:

$$h_{i,t} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij,t} W_t x_{j,t} \right) \quad (2)$$

where $\mathcal{N}(i)$ denotes the neighbor nodes of node i , α is the learnable attention score, and W is the learnable weight matrix. We adopt a multi-stage strategy for GAT, with each of the N stages comprising M layers of GAT and incorporating residual connections between stages.

Then, updated node features are concatenated following the edges to get the human-object interaction features for two hands:

$$H_{hand,t} = \bigoplus_{i \in \mathcal{N}(hand)} h_{i,t}, \text{ where } hand \in \{lh, rh\}. \quad (3)$$

Finally, we concatenate the updated hand-object interaction features with the scene feature to form the frame features for two hands:

$$F_{hand,t} = H_{hand,t} \bigoplus x_{sce,t}, \text{ where } hand \in \{lh, rh\}. \quad (4)$$

c) *Temporal propagation and action prediction*: In this phase, a TCN is used to capture long dependencies of frame features across the entire video, enabling frame-wise action prediction. Given the video frame features of the left hand,

$$F_{lh} = \{F_{lh,1}, F_{lh,2}, \dots, F_{lh,T}\}$$

and the right hand,

$$F_{rh} = \{F_{rh,1}, F_{rh,2}, \dots, F_{rh,T}\}$$

we apply a TCN block, as referenced in [14], to update these features to $\hat{F}_{lh}$ and $\hat{F}_{rh}$. The TCN block comprises L layers of 1D dilated convolution networks, with the dilation factor doubling at each successive layer. After the final convolution

layer, two SoftMax layers are used to generate frame-wise action predictions for two hands, yielding

$$Y_{lh} = \{y_{lh,1}, y_{lh,2}, \dots, y_{lh,T}\}$$

and

$$Y_{rh} = \{y_{rh,1}, y_{rh,2}, \dots, y_{rh,T}\}$$

d) *Loss function*: As the loss function, we use a combination of classification loss, smoothing loss, and segmental edit distance loss:

$$L = L_{cls} + \lambda L_{T-MSE} + \mu L_{seg-edit} \quad (5)$$

We first use a cross-entropy loss for dual-hand action classification, as follows:

$$L_{cls} = \frac{1}{T} \sum_{hand \in \{lh, rh\}} \sum_t -\log(y_{hand,t,c}) \quad (6)$$

where $y_{hand,t,c}$ is the predicted probability for the ground truth action label c of $hand$ at time t .

Then, we employ truncated mean squared error to smooth the predictions, following Equation 7, 8, 9:

$$L_{T-MSE} = \frac{1}{TC} \sum_{hand \in \{lh, rh\}} \sum_{t,c} \tilde{\Delta}_{hand,t,c}^2 \quad (7)$$

$$\tilde{\Delta}_{hand,t,c} = \begin{cases} \Delta_{hand,t,c} & \text{if } \Delta_{hand,t,c} \leq \tau \\ \tau & \text{otherwise} \end{cases} \quad (8)$$

$$\Delta_{hand,t,c} = |\log y_{hand,t,c} - \log y_{hand,t-1,c}| \quad (9)$$

As suggested in [20], assembly action segmentation should pay more attention to segment-wise sequence accuracy. Therefore, we calculate the Levenshtein distance between the predicted segment sequence $\bar{S}_{hand}$ and the ground truth segment sequence S_{hand} for $hand \in \{lh, rh\}$, formulating the segmental edit distance loss:

$$L_{seg-edit} = \frac{1}{TC} \sum_{hand \in \{lh, rh\}} \text{Levenshtein}(\bar{S}_{hand}, S_{hand}) \quad (10)$$

We consolidate consecutive and identical action labels into a single segment to obtain $\bar{S}_{hand}$ and S_{hand} .

IV. EXPERIMENTS

A. Dataset

The proposed DuHa algorithm was assessed on HA-ViD, a human assembly video dataset. We chose HA-ViD because it contains detailed dual-hand action annotations and covers a wide range of common assembly actions and parts in real-world industrial scenarios. We selected a subset of HA-ViD that includes both object bounding boxes and action labels. The selected dataset comprises 116 videos captured from three views (side, front, and top), including 219 atomic action classes. The average length of each video is approximately 1.5 minutes. For evaluation, the dataset is split into a training set of 98 videos and a testing set of 18 videos.

B. Implementation details

In our spatial propagation process, four stages are implemented, and each stage contains four layers of GAT. Object locations are determined using ground truth bounding box labels, while pre-extracted I3D features [32] are taken as the scene features. During temporal propagation, the TCN block consists of three dilated convolution layers with a progressively doubling dilation factor and a dropout rate of 0.2 after each layer. For the loss function, parameters τ , λ and μ are set to 4, 0.15, and 1, respectively. The Adam optimiser is employed with a learning rate set at 0.0001, and the total number of epochs is set to 100.

C. Comparison with the state-of-the-art

We compare the proposed DuHa with the state-of-the-art methods, including MS-TCN [14], DTGRM [17], ASRF [15], and C2F-TCN [29] on the subset of HA-ViD. The results are presented in Table I, where we report the frame-wise accuracy (Acc), segmental edit distance (Edit), and segmental F1 score at overlapping thresholds of 10%, 25%, and 50% (F1@{10, 25, 50}). From Table 1, it can be seen that DuHa outperforms other methods markedly in segmenting both hand actions. Significantly, our method employs a single model to simultaneously segment dual-hand actions, in contrast to other approaches which require separate models for each hand. Therefore, our method has stronger practicality in real-world applications that require dual-hand action understanding. Table I reveals a significant discrepancy in DuHa's performance between right and left hand action segmentation, with right hand performance notably inferior. This disparity is primarily due to the HA-ViD dataset's predominance of right-handed participants, leading to more intricate right-handed actions.

D. Ablation study

In this section, we examine the impacts of missing node compensation mechanism, scene feature inclusion, segmental edit distance loss and hand-object interaction features. All experiments are conducted over the same number of total epochs. Table II presents the results. As Table II shows, the missing node compensation mechanism and segmental edit distance loss enhance the algorithm's efficacy, particularly reflected by improvements in the Edit metric, affirming their utility in achieving higher segment-level sequence accuracy. Omitting scene features results in a notable decline in performance, especially for right-hand actions. As the complexity of right-hand actions is higher, it confirms the essential contribution of scene features to segmenting complex actions. This is because, compared with the focused but sparse hand-object interaction features, scene features capture global, dense motion information, which aids in distinguishing short-term and similar actions. Moreover, the reduced performance without hand-object interaction features highlights their crucial role in assembly action segmentation and validates the effectiveness of the proposed feature extraction method.

TABLE I
COMPARISON WITH THE STATE-OF-THE-ART ON A SUBSET OF HA-ViD.

Method	View	Left hand atomic action					Right hand atomic action					Single model for dual-hand action
		Acc	Edit	F1@{10, 25, 50}			Acc	Edit	F1@{10, 25, 50}			
MS-TCN [14]	Side	42.2	26.17	28.66	24.39	12.2	17.37	4.98	6.64	4.65	2.66	✗
	Front	41.28	31.65	28.63	21.89	12.63	16.75	5.01	6.63	4.97	3.31	
	Top	42.79	27.45	28.11	27.03	14.59	18.07	6.63	8.47	6.51	4.56	
DTGRM [17]	Side	41.81	30.39	31.2	23.4	14.89	18.34	18.94	16.78	13.05	6.99	✗
	Front	44.41	32.58	30.29	22.86	9.14	27.59	25.57	29.21	21.07	10.72	
	Top	44.04	28.34	28.99	27.03	14.59	27.57	23.24	26.09	20.79	14.37	
ASRF [15]	Side	25.78	22.16	18.18	12.65	5.53	11.88	15.33	16.39	13.58	7.96	✗
	Front	14.47	18.47	18.7	11.43	3.12	16.79	23.01	24.14	18.51	10.46	
	Top	16.94	20.41	15.81	10.31	5.5	22.44	23.34	24.62	21.98	13.19	
C2F-TCN [29]	Side	51	9.27	12.9	11.61	5.16	20.47	10.85	9.7	7.88	4.24	✗
	Front	46.62	8.11	10.9	8.48	6.06	26.32	15.02	16.96	14.96	6.98	
	Top	56.45	11.13	11.35	8.51	2.84	24.8	11.62	12.32	9.97	5.87	
DuHa(ours)	Side	50.22	35.79	40.27	34.51	23.13	34.29	33.55	36.79	30.57	23.46	✓
	Front	52.43	31.22	33.65	29.91	15.37	30.12	30.33	32.6	27.3	18.08	
	Top	52.43	29.33	34.09	28.31	17.07	30	31.22	34.69	29.68	19.04	

TABLE II
IMPACTS OF MISSING NODE COMPENSATION MECHANISM (NC), SEGMENTAL EDIT DISTANCE LOSS (SEG-EDIT), SCENE FEATURE (SF) AND HAND-OBJECT INTERACTION FEATURE (HOF) ON DUHA.

Method	View	Left hand atomic action					Right hand atomic action				
		Acc	Edit	F1@{10, 25, 50}			Acc	Edit	F1@{10, 25, 50}		
DuHa	Side	50.22	35.79	40.27	34.51	23.13	34.29	33.55	36.79	30.57	23.46
	Front	52.43	31.22	33.65	29.91	15.37	30.12	30.33	32.6	27.3	18.08
	Top	52.43	29.33	34.09	28.31	17.07	30	31.22	34.69	29.68	19.04
w/o NC	Side	50.92	30.28	35.57	30.87	19.36	34.24	33.68	38.56	33.93	22.68
	Front	55.56	32.08	36.75	32.98	18.78	28.5	27.08	31.46	26.49	15.45
	Top	50.5	25.57	29.53	25.22	15.45	30.82	27.13	32.61	27.78	19.65
w/o SF	Side	56.66	21.14	19.04	13.77	7.64	20.04	12.34	9.66	6.65	3.39
	Front	52.7	31.41	32.98	25.82	12.94	28.27	26.86	30.2	24.13	15.6
	Top	49.55	21.83	18	14.61	8.34	19.63	15.8	12.88	6.6	3.63
w/o seg-edit	Side	51.16	34.26	35.72	30.54	20.6	30.16	32.69	35.57	29.89	20.89
	Front	53.69	32.4	35.42	32.98	17.14	28.45	27.25	30.27	24.31	16.11
	Top	52.21	27.61	29.77	26.56	16.94	30.68	30.35	33.32	30.76	20.55
w/o HOF	Side	46.07	30.67	32.11	26.58	16.57	19.98	17.68	18.93	15.23	10.33
	Front	48.22	23.89	27.81	25.77	14.37	25.67	20.23	21.66	17.7	10.49
	Top	46.89	28.86	31.67	24.9	15.23	22.65	20.85	20.29	15.66	9.25

E. Comparison with action recognition

We previously argued that action recognition techniques fall short in meeting HRCA requirements due to their limited ability to achieve accurate segment-level sequences. To validate this, we compared our approach with UniFormerV2 [33], a leading action recognition method with superior performance on HA-ViD as reported in [20]. After training UniFormerV2 on our dataset and testing it via a sliding window method, the findings, detailed in Table III and contrasted against Table I, confirm the advantages of action segmentation techniques.

F. Discussion

Experimental results demonstrate DuHa’s superiority in dual-hand assembly action segmentation relative to other methods, but its accuracy remains unsatisfactory. This is partially due to the complexity of HA-ViD dataset, characterised by numerous action classes, a multitude of *null* actions, and highly varied assembly sequence orders. Moreover, it highlights the intrinsic difficulties of assembly action segmentation, which requires comprehensive reasoning about

the action-action and action-task relationships. A promising future research direction is the integration of large language models (LLM) into action segmentation to bolster reasoning capabilities. Furthermore, DuHa’s ability to simultaneously segment dual-hand actions promotes nuanced HRCA. Nonetheless, the absence of real-time action segmentation ability in DuHa, a essential ability for practical applications, highlights another future research direction.

V. CONCLUSIONS

This paper introduces a dual-hand action segmentation method, DuHa, for HRCA. DuHa employs GATs to extract dual hand-object interaction features per frame, and integrates these features with scene features that capture overall motion information. A TCN is then applied to capture the features’ temporal dependencies, facilitating the frame-wise action classification. This approach achieves segmenting dual-hand actions simultaneously using a single model. Experiments on the HA-ViD dataset, which represents real-world industrial assembly scenarios, demonstrate our method’s superiority over other methods. Despite its success, limitations remain in segmentation accuracy and

TABLE III
COMPARISON WITH ACTION RECOGNITION (UniFormerv2).

Method	View	Left hand atomic action					Right hand atomic action				
		Acc	Edit	F1@{10, 25, 50}			Acc	Edit	F1@{10, 25, 50}		
UniFormerv2 [33]	Side	41.91	16.24	17.49	14.05	6.79	19.4	17.35	17.33	12.06	6.23
	Front	45.26	19.5	22.39	17.92	10.35	26.2	18.87	22.21	15.84	8.84
	Top	36.21	14.45	16.84	12.49	5.28	25.12	20.86	22.67	15.97	9.08

practical deployment. Our future work includes: 1) Enhancing action segmentation accuracy by employing advanced reasoning mechanisms; 2) Developing real-time dual-hand segmentation methods using sliding windows; and 3) Integrating DuHa into a comprehensive HRCA system.

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